

Safin Mahmud

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with extensive experience in building production backend systems and AI-integrated applications. Architected scalable REST APIs using Django and FastAPI, designed PostgreSQL databases with query optimization and CTEs, and built ETL data pipelines for enterprise clients. Integrated LLM workflows including OpenAI, Anthropic Claude, and RAG retrieval systems into production. Shipped code to open-source platforms used by millions of developers (Sentry, PostHog). Master of Science in Computer Science with Distinction. Based in Toronto, ON.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Backend and APIs: Django, FastAPI, Django REST Framework, REST API Development, SOAP API Integration

Frontend: React.js, Next.js, Vue.js, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, MySQL, SQLite, Supabase, Database Optimization, Query Optimization

AI Tools: OpenAI API, Anthropic Claude API, LangChain, RAG Systems, Vector Databases, Whisper API, Ollama

DevOps and Cloud: Docker, Git, GitHub Actions, CI/CD, Vercel, Render

Enterprise and Tools: Splunk SDK, SAP ERP Integration, FrontAccounting ERP, Cursor IDE

PROFESSIONAL EXPERIENCE

Contract work

TechTrio

Jun 2024 – Nov 2025

- Integrated third-party SOAP and REST APIs for enterprise system interoperability, enabling seamless data exchange between ERP platforms and external services.
- Customized open-source FrontAccounting ERP codebase, developing custom SQL reports and new API endpoints to generate client-specific financial and inventory reporting data.

Akindi

Jan 2024 – Jun 2024

- Led production migration of Vue 2 application to Vue 3 (Akindi platform), modernizing frontend architecture, improving maintainability, and ensuring zero-downtime deployment.

Software Engineer

Dhaka, Bangladesh

TechTrio

January 2020 – July 2022

- Architected Rebate Management System (Django, Vue.js, PostgreSQL) replacing manual Excel workflows for SAP ERP client; automated monthly rebate calculations and enterprise reporting.
- Designed and implemented rule engine for dynamic rebate computations based on sales performance data, processing high-volume monthly transaction datasets.
- Built automated ETL pipeline extracting data from SAP ERP, transforming business rules, and loading into PostgreSQL to power monthly financial reporting.
- Optimized database performance through query optimization, CTEs, and materialized views; significantly reduced report generation time and improved system responsiveness.
- Led technical implementation, production deployment, and ongoing support for enterprise rebate management platform serving cross-functional business teams.

TECHNICAL PROJECTS

SOC Copilot: Security Incident Response Platform

Splunk AI Hackathon

- Built AI-powered SOC assistant for automated Splunk SPL query generation, threat analysis, and incident reporting with MITRE ATT&CK framework mappings.
- Integrated Ollama LLM with FastAPI backend and Next.js frontend; deployed containerized application on Google Cloud Platform.
- Technologies: Next.js, FastAPI, Splunk SDK, Ollama, LLM Integration, GCP. Links: [Devpost](#) | [Demo](#)

AI Interview Coach

Full-Stack AI Application

- Developed end-to-end interview preparation platform with resume upload, AI-generated technical questions via OpenAI API, and Whisper voice transcription.
- Implemented real-time feedback engine using FastAPI and React/Next.js; deployed on Vercel and Render with Supabase backend and authentication.
- Technologies: React, Next.js, FastAPI, OpenAI API, Whisper, Supabase, Vercel, Render.

OPEN SOURCE CONTRIBUTIONS

Sentry – github.com/getsentry/sentry

Merged Pull Request #110404

- Shipped production React and TypeScript fix to Sentry's explore feature, adding `!has` filter for drilling down into error data and improving query capabilities.
- Contributed to production monitoring platform used by millions of developers to track application errors, performance, and uptime.

PostHog – github.com/PostHog/posthog

Merged Pull Request #51697

- Merged React cohort criteria fix transforming flat API responses into nested group format for proper UI rendering and component behavior.
- Improved frontend component performance and usability for analytics teams processing large-scale event datasets.

EDUCATION

MSc in Computer Science (Distinction)

2022 – 2023

University of Greenwich, London, UK

Coursework: Advanced Software Engineering, Machine Learning, Cloud Computing, Distributed Systems

BSc in Computer Science and Engineering

2015 – 2019

North South University, Bangladesh

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Architecture, Machine Learning, Software Engineering

Publication

[Forecasting River Sediment Deposition through Satellite Image-Driven Unsupervised Machine Learning Techniques](#)

Remote Sensing Applications: Society and Environment, 2019